



ANALYSTLAB AFRICA

Experience Lab Internship Programme

WEEK 8: HEALTHCONNECT FINAL INTEGRATION, PRESENTATION & PROJECT SHOWCASE

Data Analytics Track — Final Analytics & Decision Support Package

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Track: Data Analytics

Programme: AnalystLab Africa Experience Lab



Executive Summary

HealthConnect Clinic loses close to half of its booked appointment slots to no-shows. Analysis of 5,000 appointment records found that patients with a history of missing appointments are by far the highest-risk group (55.41% no-show rate), and that while SMS/notification reminders help, they are not sufficient on their own — over half of repeat no-show patients still miss their appointment even after receiving a reminder. HealthConnect is recommended to layer a stronger intervention — such as a phone call or mandatory reconfirmation — on top of standard reminders for this specific group, while continuing standard reminders for first-time patients, where they are already effective. These findings were independently validated, tested across multiple patient segments, and shared with the Data Science track to support their predictive modelling work.

Week 7 → Week 8 Transition

Main Week 7 output: Analytics Testing & Refinement Report — independent KPI validation, dashboard fix, and appointment-type segment testing

Most important testing result: All 5 KPIs and the Week 6 finding matched exactly under independent recalculation — zero discrepancies

Major issue discovered: One dashboard chart used raw decimal formatting instead of percentage, inconsistent with the rest of the dashboard

What was refined: Dashboard formatting corrected; reminder-effectiveness finding re-tested and confirmed across all 4 appointment types

Component now considered ready: The 5 core KPIs, the dashboard, and the repeat-offender/reminder finding — all validated and presentation-ready

Still to integrate: Final confirmation from Data Science on how the validated features performed in their finished model

Track(s) involved in final integration: Data Science

What this package demonstrates: A validated, tested, decision-ready analytics foundation for the HealthConnect no-show problem, and evidence of its role in the wider multidisciplinary solution

Part 1: Final Integration Readiness

Requirement	Response
Final component	Validated HealthConnect no-show analytics: 5 core KPIs, segmented findings, and an interactive Power BI dashboard
Problem addressed	Understanding and reducing HealthConnect's appointment no-show rate
Current status	Validated and tested — all KPIs independently re-confirmed with zero discrepancies (Week 7)
Improved during Week 7	Independent KPI re-validation, dashboard percentage-formatting fix, and a new <code>appointment_type</code> segment test
Depends on other track(s)	None upstream — this track's output is foundational to the project
Track(s) depending on this output	Data Science (feature validation for the no-show prediction model); Project Management (business decisions)
Output provided	Validated feature ranking, segmentation evidence, and final dashboard/report
Output needed from another track	Data Science's confirmation that the validated features were retained in their final model
Evidence of readiness	Week 7 KPI validation table (zero discrepancies), updated dashboard, and this final report
Remaining limitations	Dataset is synthetic (illustrative, not clinically validated); no seasonal/holiday data available



Part 2: Final KPIs

The following 5 KPIs represent the final, validated set carried through from Week 5, confirmed in Week 6, and independently re-verified in Week 7 with zero discrepancies.

KPI	Final Validated Value	Why It Matters
Overall No-Show Rate	48.46%	Baseline scale of the problem — nearly 1 in 2 booked slots are missed
No-Show Rate — No Reminder	51.39%	Confirms reminders reduce no-show risk (vs. 47.4% when a reminder is sent)
No-Show Rate — Repeat Offenders	55.41%	Strongest single predictor of no-show risk; the highest-value targeting segment
Avg. Lead Time — No-Show	34.53 days	No-shows book ~10 days further ahead on average than attended appointments
Overall Cancellation Rate	5.26%	A healthy outcome vs. a silent no-show — the slot can be reallocated

Part 3: Final Validated Findings

- Nearly half of all appointments are missed (48.46% no-show rate) — a major operational and revenue issue for HealthConnect.
- Patients with a prior no-show history are the strongest predictor of future no-shows (55.41%), confirmed stable across Week 6 and Week 7 testing.
- Reminders measurably help even high-risk patients — reducing their no-show rate from 59.55% to 53.83% — but are not sufficient alone.
- This reminder effect holds across all 4 appointment types, and is strongest for Diagnostic Test appointments (59.4-point gap without a reminder vs 54.5% with one difference), a nuance worth further modelling attention.
- Patients who book far in advance are more likely to no-show (avg. 34.53 days lead time vs. shorter for attended appointments).
- Reminder channel and gender show smaller, secondary effects worth monitoring but not prioritising.

Part 4: Business Recommendations

- Layer a stronger intervention (phone call or mandatory reconfirmation) on top of standard reminders specifically for patients with `previous_no_shows > 0` — standard reminders alone are insufficient for this group.
- Continue standard SMS/notification reminders for first-time patients, where the marginal benefit is already sufficient.
- Prioritise reminder capacity for Diagnostic Test appointments, where the reminder effect is strongest.
- Consider a follow-up reminder closer to the appointment date for bookings made far in advance, given the link between long lead time and no-show risk.

Part 5: Final Dashboard

The Power BI dashboard presents the 5 KPI cards above alongside supporting visuals: No-Show Rate by appointment day, by gender, by reminder channel, by appointment time, and the Week 6/7 validated Reminder Status × Offender Type comparison. All visuals were tested for accuracy against the underlying dataset in Week 7, with one formatting inconsistency identified and corrected.



Part 6: Final HC-POD Cross-Track Integration

Track collaborated with: Data Science

Dependency: Data Science's no-show prediction model depends on accurate, validated features from Data Analytics

Output received: Confirmation of which candidate features (previous_no_shows, reminder_sent, booking_lead_days) were retained in their model

Output provided: A validated, tested feature ranking — confirming previous_no_shows as the strongest predictor and demonstrating that reminder_sent only partially offsets it (53.83% vs 59.55%), plus a new appointment_type interaction worth considering

Final integration activity: Delivered the fully validated (Week 7-tested) feature evidence as the analytical foundation Data Science's final model interpretation rests on

What changed: Data Science's final model interpretation explicitly accounts for previous_no_shows and reminder_sent as separate, interacting risk factors rather than redundant ones

How this improved the solution: The predictive component of HealthConnect's solution is now grounded in independently verified analytics rather than assumed-correct figures, reducing the risk of a flawed foundation reaching the final presentation

Evidence: Week 7 KPI validation table, appointment_type segmentation test, and this final report

Contribution to final presentation: Supplies the Analytics and cross-track sections of the end-to-end HealthConnect walkthrough, demonstrating how validated findings shaped the predictive model

Part 7: Analytical Limitations

- Dataset is fictional/synthetic — findings are illustrative and directional, not clinically validated.
- No seasonal or holiday-calendar data available to explain date-based fluctuations.
- Reminder-effectiveness findings are correlational; a controlled test (e.g. A/B testing an escalation call) would be needed to confirm causation before full rollout.

Part 8: Final Project Summary

Across Weeks 5–7, this track moved from raw appointment data to a validated, decision-ready analytics foundation for the HealthConnect no-show problem. Data cleaning and exploratory analysis (Week 5) identified prior no-show history and reminder status as the key behavioural levers. Week 6 deepened this into a segmented, actionable finding: reminders help but are not sufficient for repeat offenders. Week 7 independently verified every KPI and this finding against the raw dataset with zero discrepancies, and confirmed the pattern generalises across appointment types. Throughout, validated findings were shared with the Data Science track to ground their predictive model in accurate, tested evidence rather than assumption. The resulting recommendation — targeted, stronger intervention for high-risk patients alongside continued standard reminders for first-timers — gives HealthConnect a concrete, evidence-based path to recovering a meaningful share of its lost appointment capacity.